Supplementary Quality Control and Robustness

Coverage, paired label-swap null, and adjusted-geometry sensitivity

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S3.1 Coverage and contract QC

The H5 run manifest recorded 68 source files, no bad files, finite 3D coordinates for all files, finite 9D coordinates for all files, 3D Jacobians for all files, and 9D Jacobians for 67 files. All files contained 74 windows. The primary coordinate analyses therefore used 34 pairs; the reachability surface used 33 pairs. The absence of regional, time-reference, event, and embodied-anchor surfaces precluded those analyses.

Contract layer	EU files	HYPO files	Status
3D coordinates	34	34	Available
9D coordinates	34	34	Available
3D Jacobian	34	34	Available
9D Jacobian	34	33	One unavailable HYPO file
Regional MNPS	0	0	Not exported
Embodied anchors	0	0	Not exported

Table 1: H5 capability audit from the run manifest.

S3.2 Paired label-swap null

The robustness script independently swapped euglycemia/hypoglycemia labels within each participant for 10,000 draws. It evaluated 11 prespecified stratified-coordinate and reachability endpoints. This restricted family is a supplementary robustness test and is distinct from the broader engine-level FDR families reported in the main manuscript.

Endpoint	N	Null p	Null q
m_a median	34	0.00010	0.0011
m_e median	34	0.00030	0.0015
d_n median	34	0.00040	0.0015
e_e median	34	0.0172	0.0315
e_m median	34	0.0018	0.0050
Tube effective dimension	33	0.0147	0.0315
Tube directional entropy	33	0.0282	0.0413
Tube log-volume	33	0.2409	0.2650

Table 2: Two-sided paired label-swap null. Full 11-endpoint output is in `paired_label_swap_null.csv`.

S3.3 Adjusted-geometry sensitivity

Three hypoglycemia files were marked `adjusted` by their embedded geometry contract: `sub-004`, `sub-017`, and `sub-021`. The last participant also accounts for the unavailable 9D-Jacobian/reachability pairing. Excluding these participants preserved the sign of all 11 prespecified endpoints.

Endpoint	Full delta	Sensitivity delta	Sign preserved
m_a median	−1.096	−1.043	Yes
m_e median	−0.685	−0.639	Yes
d_n median	1.014	0.862	Yes
e_e median	−0.708	−0.812	Yes
e_m median	1.024	0.708	Yes
Tube effective dimension	0.251	0.237	Yes
Tube directional entropy	0.088	0.080	Yes
Tube log-volume	0.395	0.450	Yes

Table 3: Directional sensitivity excluding the three adjusted-geometry hypoglycemia sessions. This is not a re-estimated restricted-sample significance test.

S3.4 Reviewer-facing robustness summary

Robustness summary

The paired condition association is supported by complete coordinate coverage, a paired label-swap null, and preservation of direction after removing the three adjusted-geometry hypoglycemia files. The evidence does not support a broad reachability-collapse claim, does not resolve the pooled-anchor limitation, and cannot test glucose dose, recovery, awareness, regional anatomy, or embodied-anchor mechanisms.

Bibliography